

Spotify Customer Churn Analysis

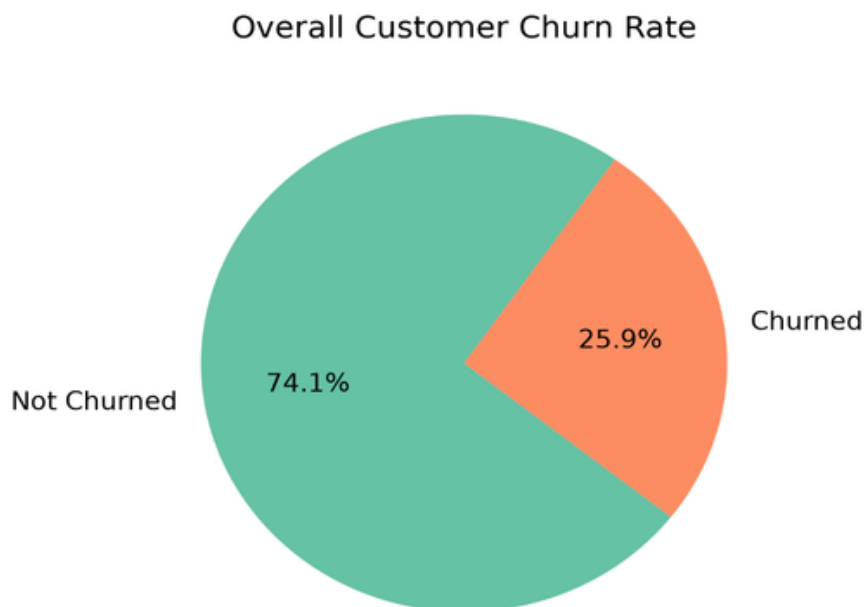
OBJECTIVE

The objective of this project is to analyze customer churn in Spotify using Python and data visualization techniques. The analysis aims to identify the overall churn rate, compare churn across different subscription plans, and examine whether offline listening is associated with customer retention.

Dataset Overview

The dataset contains user information such as subscription type, listening type, songs played per day, device type, offline listening, advertisements listened per week, and churn status. These features are analyzed to understand customer behavior and identify factors associated with churn.

Chart 1 : Overall Customer Churn Analysis



Insights

- The Family subscription plan has the highest churn rate (27.52%) among all subscription types.
- The Student plan follows with a churn rate of 26.19%, while the Free and Premium plans show slightly lower churn rates.
- The differences in churn rates are relatively small, suggesting that subscription type alone is not the only factor influencing customer churn.
- Spotify should further investigate the Family plan to understand the reasons behind its comparatively higher churn rate.
- Additional customer engagement strategies can be developed for subscription plans with higher churn.

Chart 2 : Churn Rate by Subscription Type

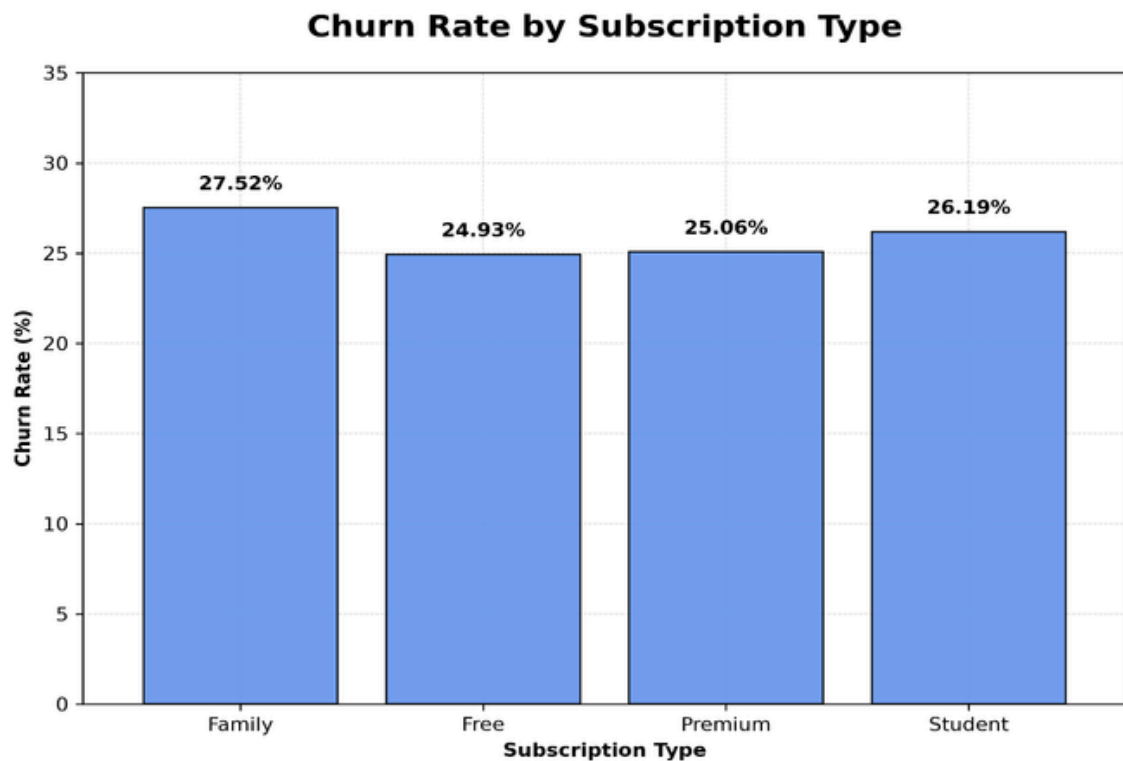


Figure 2

Insight

- The Family subscription plan records the highest churn rate at 27.52%.
- The Student plan has the second-highest churn rate at 26.19%.
- The Premium and Free plans have similar churn rates of approximately 25%.
- Although the differences between the subscription plans are not very large, the Family plan consistently shows the highest customer attrition.
- This suggests that Spotify should further examine the Family subscription to understand customer expectations, pricing, or feature usage.
- Subscription type alone does not completely explain churn, indicating that customer behaviour and service features should also be considered.
- Combining subscription-based analysis with behavioural analysis provides a better understanding of customer retention.

Chart 3 : Offline Listening vs Customer Churn

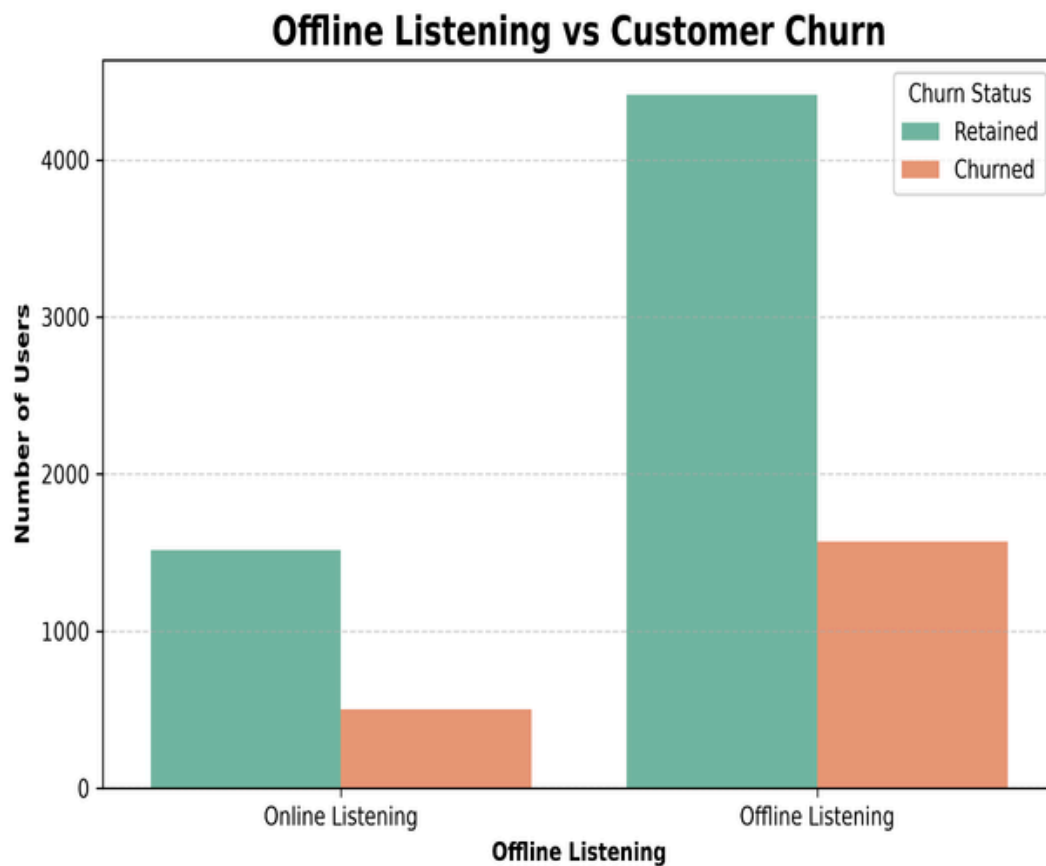


Figure 3

Insight

- Users who utilize the offline listening feature have a higher number of retained customers than churned customers.
- This suggests a positive association between offline listening and customer retention.
- Offline listening allows users to enjoy uninterrupted music without an internet connection, which may improve overall user experience.
- Customers who regularly use platform features are often more engaged with the service.
- Although this chart does not prove that offline listening directly reduces churn, it indicates that feature usage may influence customer retention.
- Encouraging users to explore and adopt offline listening could contribute to improving customer engagement and loyalty.
- Additional behavioural features can be analyzed in future studies to better understand customer churn patterns.

Conclusion

The analysis of the Spotify customer churn dataset provides valuable insights into customer retention patterns. The overall churn rate of 25.9% indicates that a considerable proportion of users discontinue using the platform. Among the subscription plans, the Family plan experiences the highest churn rate, suggesting that this category may require additional attention. The analysis also shows that users who utilize the offline listening feature are retained in greater numbers than those who churn. While these findings do not establish a direct cause-and-effect relationship, they highlight important behavioural patterns that Spotify can consider when designing customer retention strategies. Overall, data visualization proved effective in identifying meaningful trends and supporting business decision-making.

Recommendations

- Promote the offline listening feature through in-app notifications and awareness campaigns.
- Conduct further analysis of the Family subscription plan to identify the causes of its higher churn rate.
- Develop personalized retention strategies for customers showing signs of disengagement.
- Continuously monitor customer behavior and churn trends to improve long-term retention.
- Use data-driven insights to enhance user experience and increase customer satisfaction.